

```
mysql> CREATE TABLE Bank_Transaction (
-> Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
-> Account_No INT,
-> Transaction_Type VARCHAR(20),
-> Amount DECIMAL(12,2),
-> Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
-> FOREIGN KEY (Account_No)
-> REFERENCES Account(Account_No)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Customer
-> (Customer_ID, Customer_Name, Phone, City)
-> VALUES
-> (101, 'Ravi Kumar', '9876543210', 'Hyderabad'),
-> (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
-> (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
-> (104, 'Sneha Rao', '9876543213', 'Chennai'),
-> (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Account
-> (Account_No, Customer_ID, Account_Type, Balance, Branch)
-> VALUES
-> (10001, 101, 'Savings', 50000, 'Hyderabad'),
-> (10002, 102, 'Savings', 75000, 'Vijayawada'),
-> (10003, 103, 'Current', 120000, 'Bangalore'),
-> (10004, 104, 'Savings', 45000, 'Chennai'),
-> (10005, 105, 'Current', 90000, 'Hyderabad');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Bank_Transaction
-> (Account_No, Transaction_Type, Amount)
-> VALUES
-> (10001, 'DEPOSIT', 10000),
-> (10002, 'DEPOSIT', 15000),
-> (10003, 'WITHDRAW', 20000),
-> (10004, 'DEPOSIT', 5000),
-> (10005, 'WITHDRAW', 10000);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Bank_Transaction
-> (Account_No, Transaction_Type, Amount)
-> VALUES
-> (10001, 'DEPOSIT', 10000),
-> (10002, 'DEPOSIT', 15000),
-> (10003, 'WITHDRAW', 20000),
-> (10004, 'DEPOSIT', 5000),
-> (10005, 'WITHDRAW', 10000);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Customer;
+-----+-----+-----+-----+
| Customer_ID | Customer_Name | Phone | City |
+-----+-----+-----+-----+
| 101 | Ravi Kumar | 9876543210 | Hyderabad |
| 102 | Priya Sharma | 9876543211 | Vijayawada |
| 103 | Arjun Reddy | 9876543212 | Bangalore |
| 104 | Sneha Rao | 9876543213 | Chennai |
| 105 | Kiran Kumar | 9876543214 | Hyderabad |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> CREATE DATABASE Bank_ACID_DB;
Query OK, 1 row affected (0.01 sec)

mysql> USE Bank_ACID_DB;
Database changed

mysql> CREATE TABLE Customer (
-> Customer_ID INT PRIMARY KEY,
-> Customer_Name VARCHAR(100) NOT NULL,
-> Phone VARCHAR(15),
-> City VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Account (
-> Account_No INT PRIMARY KEY,
-> Customer_ID INT,
-> Account_Type VARCHAR(20),
-> Balance DECIMAL(12,2),
-> Branch VARCHAR(50),
-> FOREIGN KEY (Customer_ID)
-> REFERENCES Customer(Customer_ID)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 17:01:26
2	10002	DEPOSIT	15000.00	2026-08-15 17:01:26
3	10003	WITHDRAW	20000.00	2026-08-15 17:01:26
4	10004	DEPOSIT	5000.00	2026-08-15 17:01:26
5	10005	WITHDRAW	10000.00	2026-08-15 17:01:26
6	10001	DEPOSIT	10000.00	2026-08-15 17:01:31
7	10002	DEPOSIT	15000.00	2026-08-15 17:01:31
8	10003	WITHDRAW	20000.00	2026-08-15 17:01:31
9	10004	DEPOSIT	5000.00	2026-08-15 17:01:31
10	10005	WITHDRAW	10000.00	2026-08-15 17:01:31

10 rows in set (0.00 sec)

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 5000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> SELECT * FROM Account
  -> WHERE Account_No = 10001;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

1 row in set (0.00 sec)

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 10000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

A **transaction** is a group of SQL operations treated as one unit.

- START TRANSACTION / BEGIN → Starts a transaction.
- COMMIT → Permanently saves changes.
- ROLLBACK → Cancels changes made in the transaction.
- SAVEPOINT → Creates a point to which you can partially roll back.
- ROLLBACK TO SAVEPOINT → Cancels changes after a particular savepoint.
- RELEASE SAVEPOINT → Removes a savepoint.

```
mysql>
mysql> SELECT * FROM Account
  -> WHERE Account_No = 10001;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
|      10001 |          101 | Savings      | 45000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 5000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT Deposit1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 3000
  -> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> SAVEPOINT Withdrawal1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 10000
  -> WHERE Account_No = 10003;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SAVEPOINT Withdrawal1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 5000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT Deposit1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 3000
```

```
mysql> UPDATE Account
-> SET Balance = Balance - 3000
-> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> SAVEPOINT Withdrawal1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE Account
-> SET Balance = Balance + 10000
-> WHERE Account_No = 10003;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> ROLLBACK TO SAVEPOINT Withdrawal1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE Account
-> SET Balance = Balance - 10000
-> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> UPDATE Account
```

```
mysql>
mysql> UPDATE Account
-> SET Balance = Balance + 10000
-> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT @@SESSION.transaction_isolation;
+-----+
| @@SESSION.transaction_isolation |
+-----+
| REPEATABLE-READ                  |
+-----+
1 row in set (0.02 sec)
```

```
mysql> SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE Account
```

```
mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 5000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 3000
  -> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 10000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```

```
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 15000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 15000
  -> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 20000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```

```
mysql> SAVEPOINT AfterDebit;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 20000
  -> WHERE Account_No = 10003;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 5000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 2000
  -> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT BeforeThirdUpdate;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 10000
  -> WHERE Account_No = 10003;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SAVEPOINT BeforeThirdUpdate;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 10000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 10000
  -> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)
```

```

Query OK, 0 rows affected (0.02 sec)

mysql> SELECT SUM(Balance) AS Total_Balance
  -> FROM Account;
+-----+
| Total_Balance |
+-----+
|      394000.00 |
+-----+
1 row in set (0.01 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance - 10000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 10000
  -> WHERE Account_No = 10002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> SELECT SUM(Balance) AS Total_Balance
  -> FROM Account;

```

```

mysql>
mysql> SELECT SUM(Balance) AS Total_Balance
  -> FROM Account;
+-----+
| Total_Balance |
+-----+
|      394000.00 |
+-----+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 5000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> SELECT *
  -> FROM Account
  -> WHERE Account_No = 10001;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
|      10001 |          101 | Savings      | 15000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *

```

```

mysql> SELECT *
  -> FROM Account
  -> WHERE Account_No = 10001;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
|      10001 |          101 | Savings      | 15000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 20000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> SELECT Balance
  -> FROM Account
  -> WHERE Account_No = 10001;
+-----+
| Balance |
+-----+
| 35000.00 |
+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT Balance
  -> FROM Account
  -> WHERE Account_No = 10001;
+-----+
| Balance |
+-----+
| 35000.00 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Account
  -> SET Balance = Balance + 10000
  -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

```



```
mysql>
mysql> ROLLBACK TO SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @@SESSION.transaction_isolation;
+-----+
| @@SESSION.transaction_isolation |
+-----+
| SERIALIZABLE                     |
+-----+
1 row in set (0.00 sec)

mysql> SELECT @@GLOBAL.transaction_isolation;
+-----+
| @@GLOBAL.transaction_isolation |
+-----+
| REPEATABLE-READ                 |
+-----+
1 row in set (0.00 sec)

mysql> SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @@SESSION.transaction_isolation;
+-----+
| @@SESSION.transaction_isolation |
+-----+
| SERIALIZABLE                     |
+-----+
1 row in set (0.00 sec)

mysql> |
```